Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/06/2020
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	GeneLab: Extending the Reach of the GeneLab Data System Across Kingdoms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	12/17/2018	**End Date:**	12/16/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/16/2021 per NSSC information (Ed., 12/30/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin System)		
Grant/Contract No.:	80NSSC19K0132		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The GeneLab data repository is a rapidly expanding and remarkably powerful resource with which to dissect organismal responses to spaceflight at levels ranging from the metagenome to the transcriptome, and proteome. The research in this GeneLab Data System study seeks to extend the bioinformatics tool development for GeneLab data to a cross-kingdom architecture. Making such inter-organismal comparisons possible will facilitate mining across the wealth of datasets within the GeneLab system and drive insights into the fundamental responses to spaceflight shared by biological systems as a whole. This work approaches some of GeneLab's core questions as to how implementation of large scale dataset analyses can help produce new insights and hypotheses about the effects of spaceflight on biology. We have developed an interactive data visualization and mining environment based on the Qlik engine, which covers all the biology data deposited within GeneLab. This approach allows users to make interactive comparisons and filters between datasets to mine for both common and unique features of the transcriptomes of the GeneLab data. Current work is focused on maintaining and expanding this data environment and moving towards a highly intuitive user interface. This tool will allow users to rapidly move from data exploration to identifying spaceflight-response genes and from there to testable hypotheses. Specific aims: (1) Import the diversity of the datasets from the GeneLab Data repository into the Qlik database and develop Orthologous Matrix- and Gene Ontology-driven approaches to translate between them in a kingdom-transparent manner. (2) Develop common phenotyping tools and approaches for the research community to facilitate the testing of potential roles of these genes in spaceflight-related factors. This effort will focus on elements of spaceflight such as microgravity and hypoxic stress response that are applicable to a wide range of organisms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is to develop an in silico analysis environment to explore GeneLab's rich set of spaceflight and spaceflight-related data. This approach should help uncover not only patterns of 'omic-level responses to spaceflight but also provide insight into how biology reacts to the range of stimuli related to spaceflight such as to gravity, radiation, and mechanical forces on Earth. The database allows comparisons across Kingdoms, providing a new tool for biologists to also ask how universal the molecular responses they are characterizing are across the diversity of life on Earth.
Task Progress:	The Test Of Arabidopsis Space Transcriptome (TOAST) database is a web-based data visualization and analysis environment for all publicly available plant science spaceflight data. The system imports data from GeneLab, from the original publications on each study, and from datasets made available by the primary researchers. These data have then been manually curated and linked in a relational database that allows researchers to rapidly compare transcriptome patterns across studies of interest. This tool has been through beta testing and optimization and the fully functional version is publicly available at: https:// TOAST imports the published primary omics analysis from the original researchers for each project and then uses these data to perform comparisons. This approach has the advantage of capitalizing on the insights these researchers have about their own experiments and how they have approached their omics work. The drawback is that this aggregates data from different researchers who have used multiple different analytical pipelines to generate, e.g., their lists of differentially expressed genes. In a complementary approach, GeneLab has developed standardized pipelines for analysis of microarray and RNAseq data and is running and depositing these results for each dataset in the GeneLab data repository. This approach should make comparisons between studies more robust. Therefore, to help understand these standardized comparisons and especially how they relate to those of the original researchers, a curated metadata matrix has been developed and cross-referenced to the relevant GeneLab analyses. The combination of this matrix and the standardized analyses has then allowed the asking of which variables affect the spaceflight data most significantly, so called Batch Effects. A major goal for this work is to extend the data visualizations described above to all the data available in the GeneLab data repository, including animal, plant, and microbial datasets. This has required a method to map equivalent genes (orthologs) across species. Working with the Orthologous Matrix consortium the ortholog comparison tables for the organisms within GeneLab datasets have been imported and visualization and cross-referencing tools generated to make these comparisons between the diversity of GeneLab datasets transparent to the user. The web app to access this data engine (the X-species Explorer) is undergoing optimization of its user interface. In a parallel effort to help space biologists explore the effects of radiation across the available spaceflight omics-level data, a data visualization environment called the Rad-Bio-App has also been developed. In collaboration with GeneLab, all the available radiation dosage data from the spaceflight experimentation deposited in GeneLab, as well as that from ground-based studies that have used a variety of sources and accelerators have been aggregated and manually curated. All this data is then linked in a database to the associated GeneLab data accessions. The App then allows users to explore the GeneLab datasets for their associated radiation environments and filter for similar and different experiment parameters.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Barker R, Lombardino J, Rasmussen K, Gilroy S. "Test of Arabidopsis Space Transcriptome: A discovery environment to explore multiple plant biology spaceflight experiments." Front Plant Sci. 2020 Mar 4;11:147. https://doi.org/10.3389/fpls.2020.00147 ; PMID: 32265943 ; PMCID: PMC7076552 , Mar-2020
Articles in Peer-reviewed Journals	Rutter L, Barker R, Bezdan D, Cope H, Costes SV, Degoricija L, Fisch KM, Gabitto MI, Gebre S, Giacomello S, Gilroy S, Green SJ, Mason CE, Reinsch SS, Szewczyk NJ, Taylor DM, Galazka JM, Herranz R, Muratani M. "A new era for space life science: International standards for space omics processing." Patterns (N Y). 2020 Dec 11;1(9):100148. https://doi.org/10.1016/j.patter.2020.100148 ; PMID: 33336201; PMCID: PMC7733874 , Dec-2020